



CERTIFICATE OF ANALYSIS

Natural gas composition and calorific value: 2025 quarterly samples

Report number	SAL-25-GC-0412
Date of issue	28 November 2025
Client	Dunes Energy Company, Eastern Dunes Field Central Processing Facility 1 (CPF-1)
Methods	Composition by gas chromatography, ISO 6974-5; calorific value, density and carbon content calculated per ISO 6976:2016
Reference conditions	Combustion 15 °C, metering 15 °C, 101.325 kPa; ideal molar volume 23.645 m3/kmol
Expanded uncertainty (k = 2)	NCV ±0.3%; methane ±0.15 mol %; CO2 emission factor ±0.5%

Samples

Sample ID	Sampling point	Sampled	Analysed	Sampled by
SAL-25-GC-0412-FG-Q1	SP-3001 fuel gas header	11-Feb-2025	13-Feb-2025	SAL field technician
SAL-25-GC-0412-FL-Q1	SP-5101 HP flare header	11-Feb-2025	13-Feb-2025	SAL field technician
SAL-25-GC-0412-FG-Q2	SP-3001 fuel gas header	13-May-2025	15-May-2025	SAL field technician
SAL-25-GC-0412-FL-Q2	SP-5101 HP flare header	13-May-2025	15-May-2025	SAL field technician
SAL-25-GC-0412-FG-Q3	SP-3001 fuel gas header	12-Aug-2025	14-Aug-2025	SAL field technician
SAL-25-GC-0412-FL-Q3	SP-5101 HP flare header	12-Aug-2025	14-Aug-2025	SAL field technician
SAL-25-GC-0412-FG-Q4	SP-3001 fuel gas header	11-Nov-2025	13-Nov-2025	SAL field technician
SAL-25-GC-0412-FL-Q4	SP-5101 HP flare header	11-Nov-2025	13-Nov-2025	SAL field technician

Results: fuel gas (SP-3001)

Component (mol %)	Q1 11-Feb-2025	Q2 13-May-2025	Q3 12-Aug-2025	Q4 11-Nov-2025	Mean
Methane	89.33	89.35	89.46	89.10	89.31
Ethane	5.73	5.99	5.72	6.24	5.92
Propane	1.54	1.53	1.55	1.37	1.50
i-Butane	0.34	0.27	0.20	0.22	0.26
n-Butane	0.40	0.29	0.40	0.43	0.38
Pentanes plus (as n-C5)	0.16	0.21	0.13	0.13	0.16
Carbon dioxide	1.38	1.26	1.42	1.40	1.36
Nitrogen	1.12	1.10	1.12	1.11	1.11
Total	100.00	100.00	100.00	100.00	100.00

Calculated property	Q1 11-Feb-2025	Q2 13-May-2025	Q3 12-Aug-2025	Q4 11-Nov-2025	Mean
Molar mass (kg/kmol)	18.20	18.15	18.14	18.17	18.16
Relative density (air = 1)	0.6284	0.6267	0.6262	0.6275	0.6272
Net calorific value (MJ/Sm ³)	36.17	36.19	36.02	36.11	36.12
Carbon content (wt %)	72.96	73.11	72.86	72.92	72.96
CO ₂ emission factor (t CO ₂ /TJ)	56.89	56.81	56.86	56.88	56.86

Results: HP flare header gas (SP-5101)

Component (mol %)	Q1 11-Feb-2025	Q2 13-May-2025	Q3 12-Aug-2025	Q4 11-Nov-2025	Mean
Methane	77.81	78.14	77.76	77.73	77.86
Ethane	10.27	9.97	10.42	10.23	10.22
Propane	5.55	5.41	5.51	5.51	5.50
i-Butane	1.00	0.99	0.98	1.01	0.99
n-Butane	1.71	1.74	1.70	1.73	1.72
Pentanes plus (as n-C ₅)	0.87	0.93	0.88	0.91	0.90
Carbon dioxide	2.33	2.37	2.30	2.38	2.34
Nitrogen	0.46	0.45	0.45	0.50	0.47
Total	100.00	100.00	100.00	100.00	100.00

Calculated property	Q1 11-Feb-2025	Q2 13-May-2025	Q3 12-Aug-2025	Q4 11-Nov-2025	Mean
Molar mass (kg/kmol)	21.38	21.35	21.37	21.41	21.38
Relative density (air = 1)	0.7380	0.7370	0.7378	0.7393	0.7381
Net calorific value (MJ/Sm ³)	41.66	41.58	41.68	41.66	41.65
Carbon content (wt %)	74.46	74.41	74.50	74.39	74.44
CO ₂ emission factor (t CO ₂ /TJ)	59.20	59.20	59.19	59.24	59.21
CO ₂ per 10 ³ Sm ³ flared at CE 0.98 (t)	2.418	2.413	2.419	2.420	2.417
CH ₄ per 10 ³ Sm ³ vented (t)	0.5279	0.5302	0.5276	0.5274	0.5283

The CO₂ emission factor is calculated as the total carbon (hydrocarbon carbon plus CO₂) multiplied by 44.010 and divided by the net calorific value. Flare factors assume the stated combustion efficiency applied to hydrocarbon carbon; CO₂ present in the gas is emitted unchanged. Results relate only to the samples tested.

Fatima Zahra Idrissi

Fatima Zahra Idrissi
Analyst
28 November 2025

Tariq Suleiman

Dr. Tariq Suleiman
Laboratory Manager (approved)
28 November 2025

